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Power analysis of cell-type deconvolution across tissues

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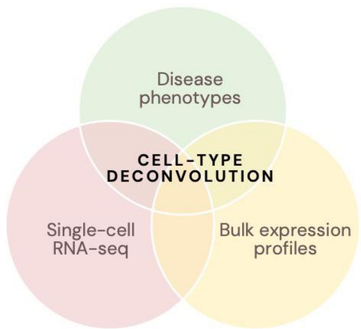


1 What is Cellular Deconvolution?

Cell-type deconvolution refers to computational methods developed to estimate the proportions of cell-types in tissue samples. Although, a great amount of bulk expression data exist in large databases, cell-type heterogeneity is masked in these data because only the average gene expression of all cells has been measured. Deconvolution methods aim to reveal the hidden layer of cell-type specific expression in tissue. Not only can this information aid downstream analyses but it can also facilitate to study the contribution of cell proportions in disease phenotypes.

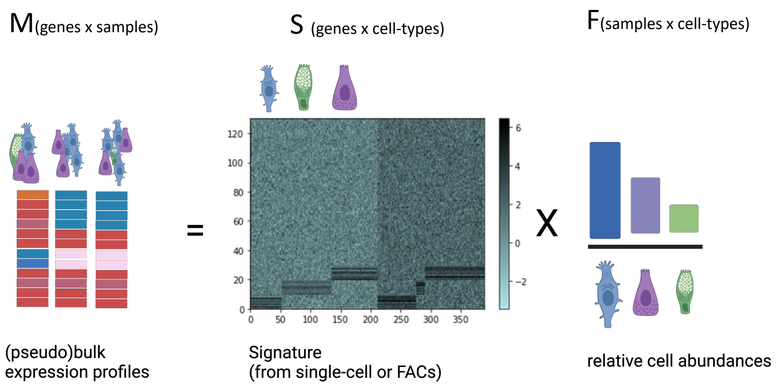


Fig 1: The deconvolution problem

Deconvolution approaches fit a linear model to explain the observed expression of a gene by building a weighted sum of cell-type specific expression profiles in a signature matrix **S** and relative fractions **F** of the cell types in the mixture **M**

2 Aim

1. Generate suitable synthetic bulk data with ground truth
2. Provide guidance for selection of deconvolution methods
3. Understand the features that affect deconvolution performance (reference, pre-processing, cell-type number)
4. Deconvolve accurately bulk tumour data from TCGA and study the contribution of cell proportions in disease phenotypes

3 Working Datasets

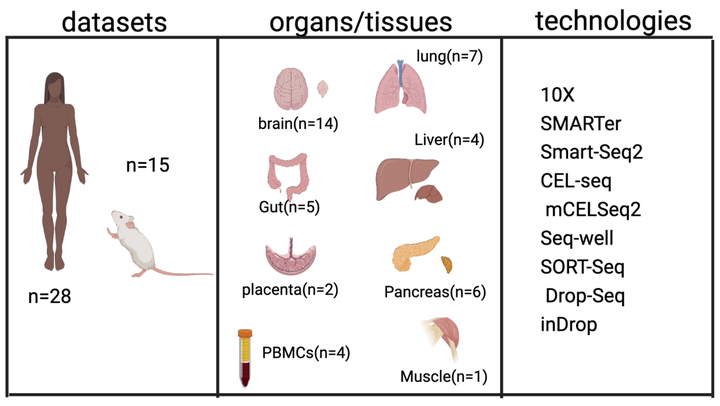
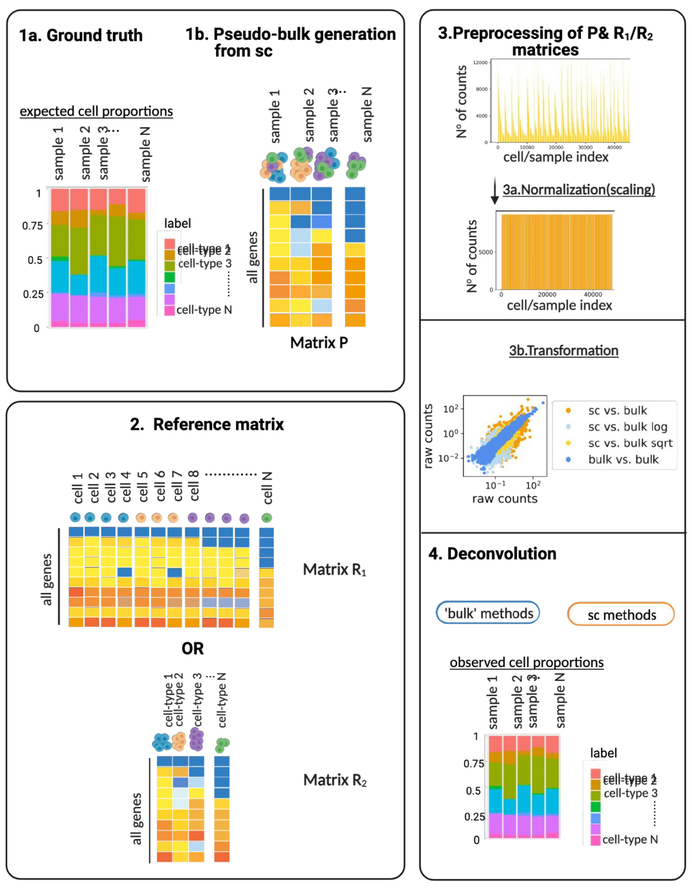


Fig 2: Brief description of single-cell RNA-seq datasets used in the study

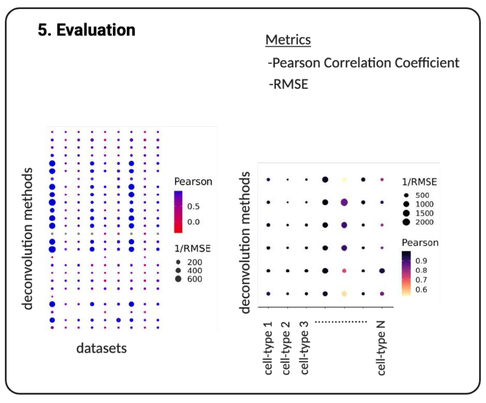
43 datasets covering 8 tissues and both human and mouse were collected, curated and utilised to generate pseudo-bulk profiles with ground truth and suitable references for the deconvolution study

4 Methodology



CATD

Pipeline & Package



Functional-Genomics/CATD
Functional-Genomics/CATD_snakemake

Fig 3: Benchmarking pipeline of cell-type deconvolution methods.

41 single-cell datasets were leveraged to construct ground truth(1a), pseudo-bulk(1b) and reference matrices(2). Resulting expression matrices can be processed by applying various transformation and/or normalization methods(3a,3b). Afterwards, pseudo-bulk profiles are deconvolved by 28 different methods and evaluated using two different metrics (R and RMSE)

5 Generating realistic pseudo-bulks

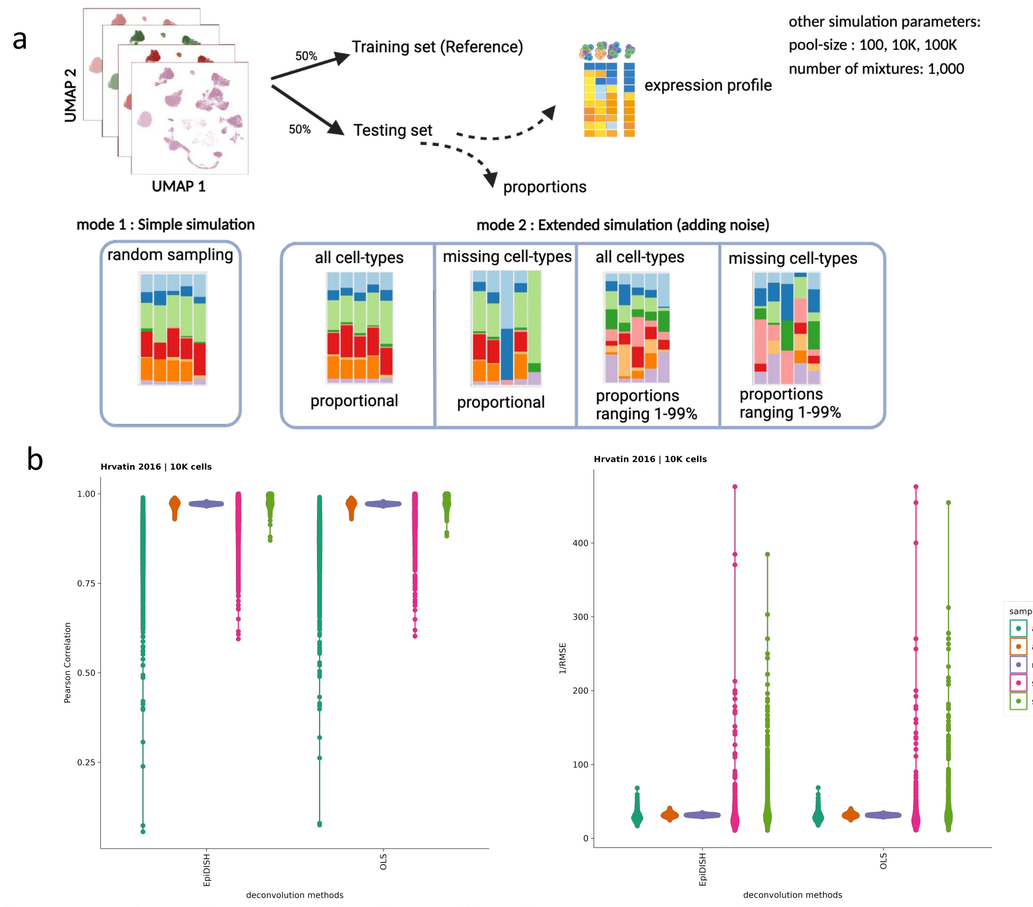


Fig 4: Construction of pseudo-bulk profiles for new method development.

a. Workflow for pseudo-bulk generation b. Performance of 2 deconvolution methods across the different sampling methods

6 Effect of the Reference

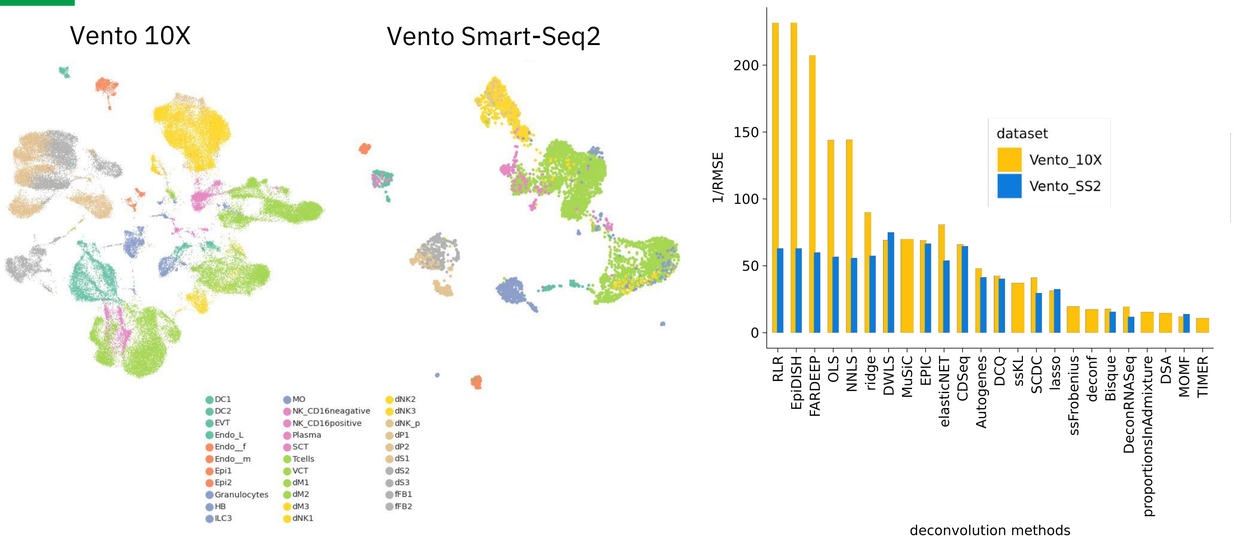


Fig 5: Effect of the Reference in deconvolution. (a)Umap plot of 2 single-cell datasets from 10X and Smart-seq2 technology. (b) Deconvolution results from pseudo-bulks created from 2 different single-cell and deconvolved with their adjacent references.

7 Large-scale evaluation

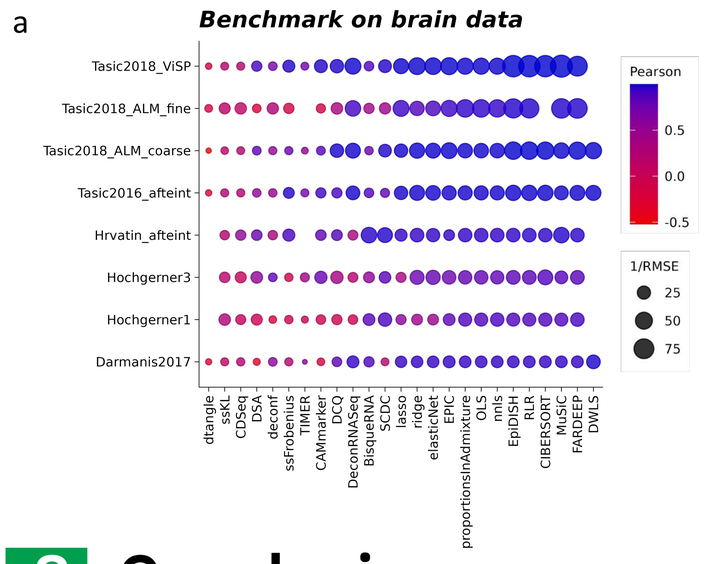


Fig 6: Large-scale evaluation of deconvolution across tissues. Dotplot(a) shows the performance of deconvolution methods across 8 different brain datasets. Pseudo-bulks and references were created from the same single-cell dataset(self-reference)

8 Conclusions

- ✓ Parameters for pseudo-bulk generation should be selected carefully to reflect real bulk data variability
- ✓ Batch effect between the reference and the pseudo-bulk result affect deconvolution performance severely
- ✓ Large –scale benchmark can identify the best performing deconvolution methods

1. Baron, M. *et al. Cell Syst.* **3**, 346-360.e4 (2016).
2. Avila Cobos et al.. *Nat. Commun.* **11**, 5650 (2020)
3. Vento-Tormo, R., Efremova, M., Botting, R.A. *et al.* . *Nature* **563**, 347–353 (2018)

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